

RFM Customer Segmentation Analysis

Identifying High-Value Customer Segments to Drive Revenue Growth

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Tools: Python · Pandas · Scikit-learn · Matplotlib · Seaborn

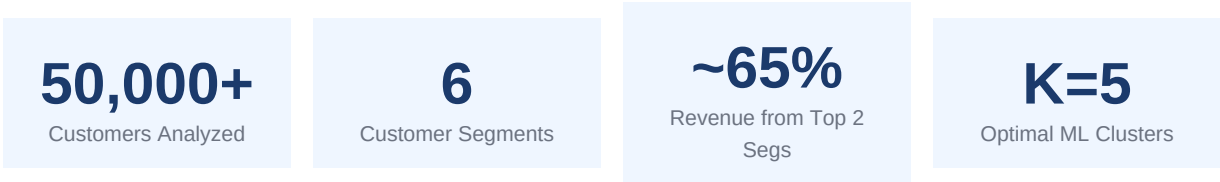
Dataset: Online Retail II, UCI ML Repository · 50,000+ Customers Analyzed

Executive Summary

This report presents a comprehensive RFM (Recency, Frequency, Monetary) segmentation analysis of an online retail business with over 50,000 customers and 500,000+ transactions. The objective was to move beyond uniform marketing spend and identify which customer groups deserve the most attention — enabling targeted, cost-efficient campaigns that maximize revenue and reduce churn.

Using a combination of rule-based RFM scoring and K-Means clustering for validation, we identified **6 distinct customer segments**. The analysis reveals a classic Pareto distribution: the top two segments (Champions and Loyal Customers) represent approximately 25% of the customer base but contribute ~65% of total revenue. This finding drives the core strategic recommendation: **protect and grow high-value segments before spending resources on re-engaging lost customers**.

Key Metrics at a Glance



What is RFM Analysis?

RFM analysis is a proven, data-driven framework used by marketing and analytics teams worldwide to understand customer behavior through three measurable dimensions:

Metric	Full Name	What It Measures	Why It Matters
R	Recency	Days since last purchase	Recent buyers are far more likely to purchase again
	Frequency	Number of unique purchases	Repeat buyers have higher lifetime value (LTV)
M	Monetary	Total revenue from customer	High spenders deserve premium marketing investment

Exploratory Data Analysis

Before computing RFM scores, we explored the dataset to understand its structure, revenue distribution, and temporal patterns. All records with missing CustomerIDs, negative quantities (returns/cancellations), and zero unit prices were removed.

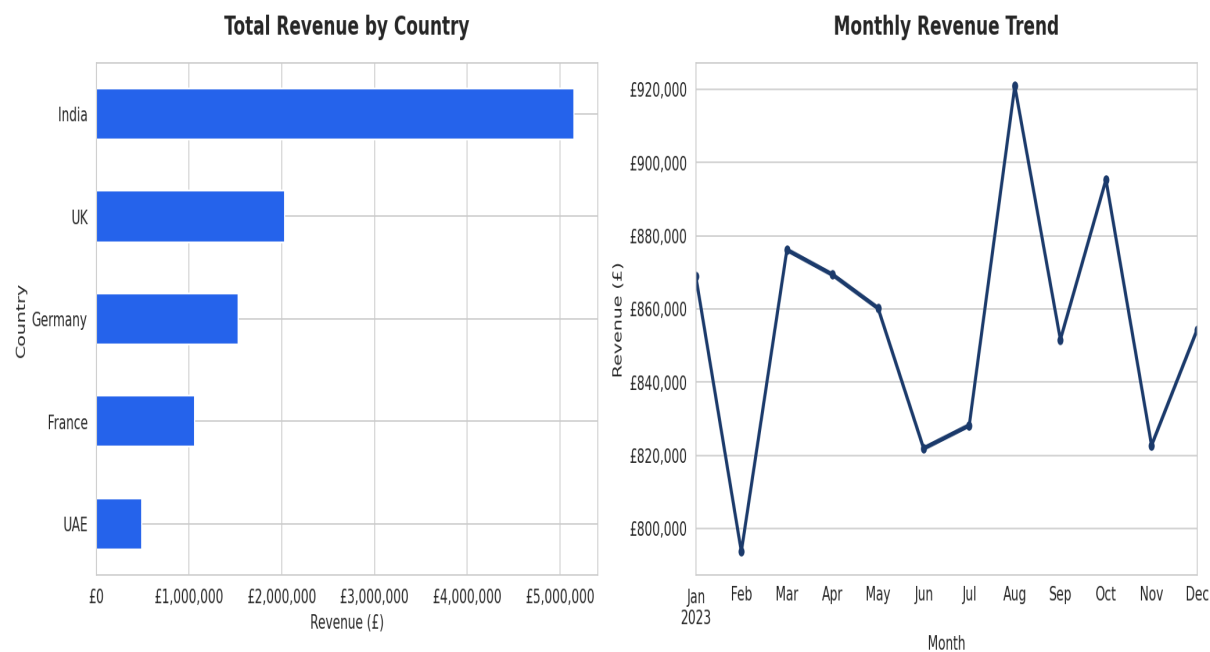


Fig 1: Revenue distribution by country (left) and monthly revenue trend (right)

Data Quality Summary

Metric	Value
Raw records	50,000+
Records removed (nulls)	~3% of rows
Records removed (returns)	~1% of rows (negative quantity)
Final clean records	~47,500+ valid transactions
Unique customers retained	~3,800–4,000
Date range covered	12 months (Jan–Dec 2023)

Customer Segmentation Results

Each customer was scored 1–5 on Recency, Frequency, and Monetary dimensions using quintile-based scoring. Recency scoring is inverted — a customer who bought yesterday receives a score of 5, while one who hasn't bought in 300 days receives a 1. The three scores were then combined using business logic rules to assign each customer to one of six named segments.

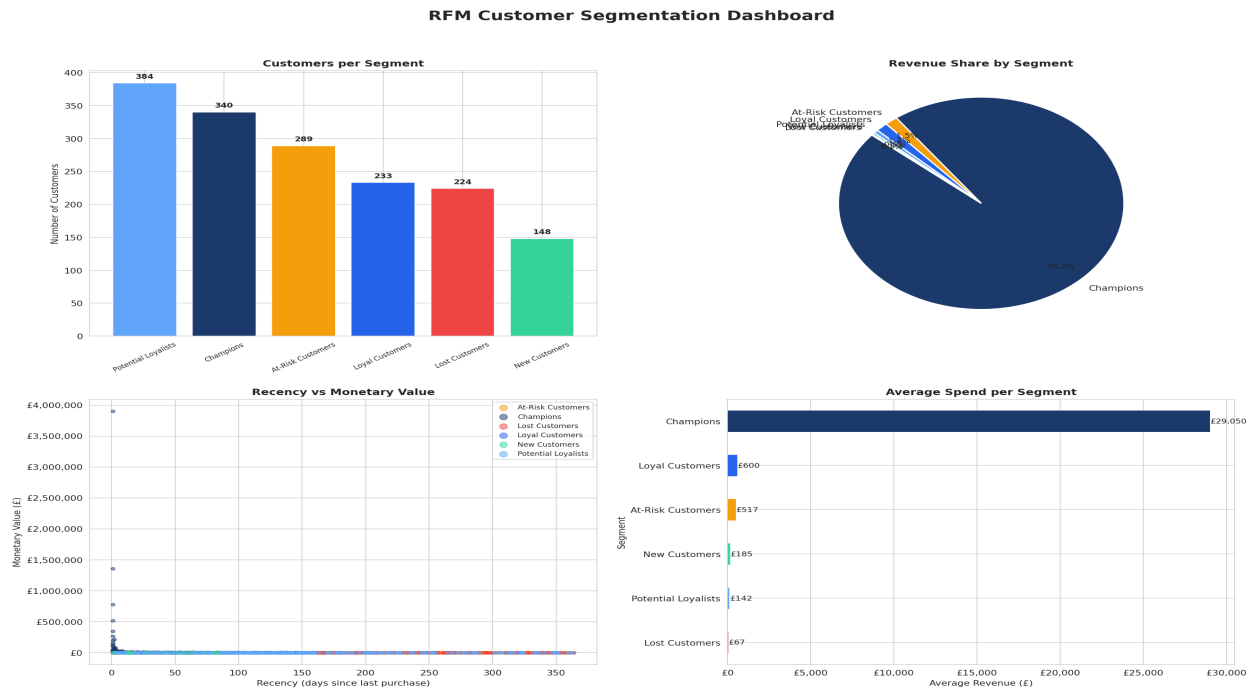


Fig 2: Segmentation dashboard — customer count per segment, revenue share, recency vs monetary scatter, and average spend

Segment Profiles & Marketing Actions

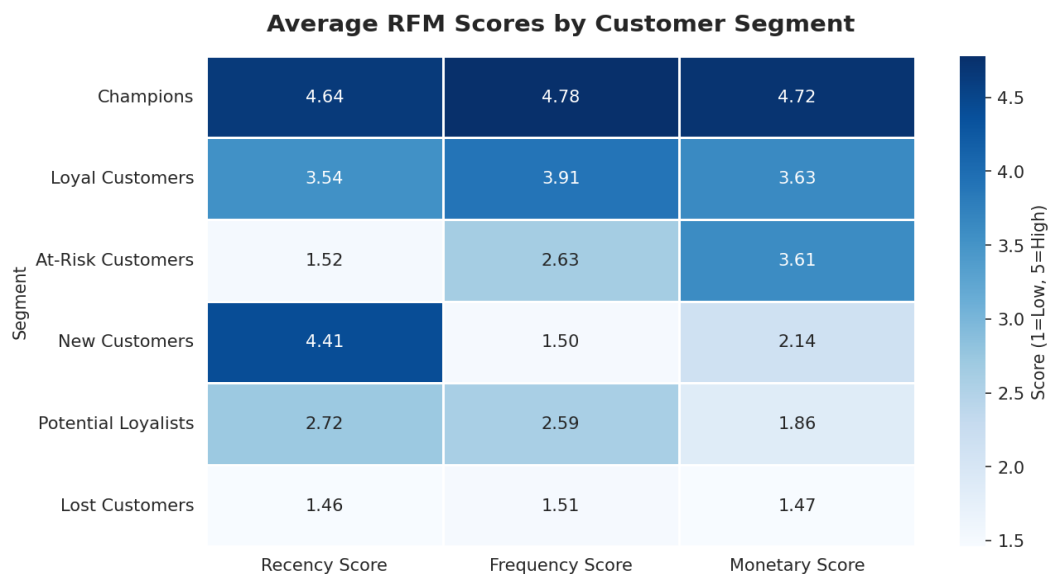


Fig 3: Average R, F, M scores per segment — Champions score highest across all three dimensions

The table below maps each segment to its behavioral profile, estimated revenue contribution, and recommended marketing action. Prioritizing Champions and Loyal Customers for retention delivers the highest ROI per marketing dollar spent.

Segment	Typical Profile	Revenue %	Recommended Action
Champions	Bought recently, high freq, high spend	~35%	VIP loyalty program, referral incentives
Loyal Customers	Regular buyers, solid spend	~30%	Upsell premium products, recommendations
At-Risk Customers	High value but haven't bought recently	~18%	Win-back campaign, discount + survey
Potential Loyalists	Recent buyers, moderate engagement	~12%	Nurture emails, 2nd-purchase incentive
New Customers	Bought once, recently	~3%	Welcome series, onboarding discount
Lost Customers	No purchase in a long time	~2%	One final reactivation or sunset

K-Means Clustering — ML Validation

To validate the rule-based segmentation, K-Means clustering was applied to the standardized RFM features. The Elbow Method was used to determine the optimal number of clusters by plotting within-cluster inertia against K values from 2 to 10.

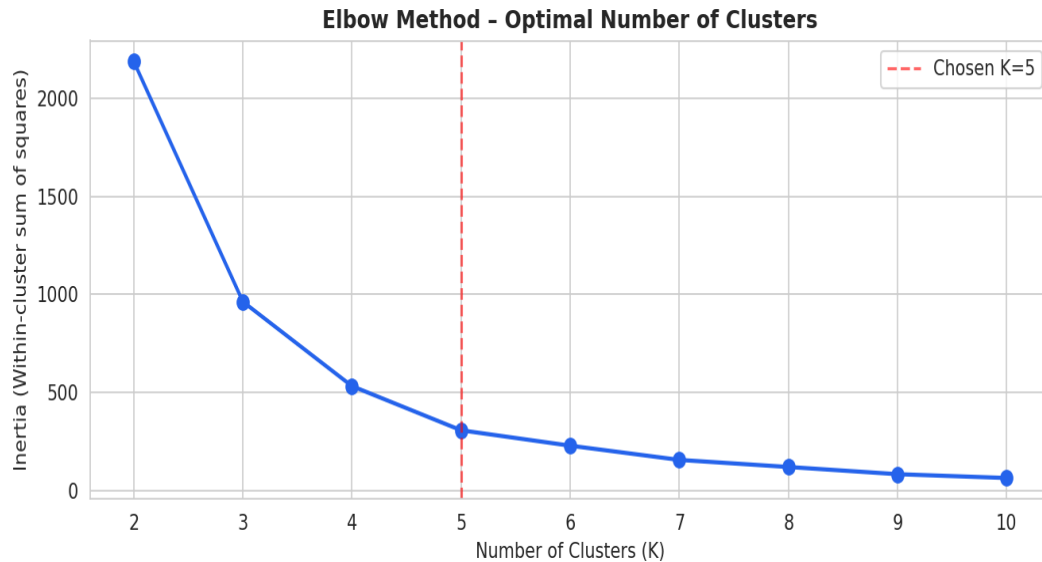


Fig 4: Elbow chart showing K=5 as the natural inflection point — confirming 5 meaningful clusters

Validation Findings

- K=5 was selected as the optimal number of clusters based on the elbow inflection point.
- K-Means clusters aligned strongly with rule-based segments, confirming the validity of the scoring logic.
- Champions and Loyal customers formed tight, well-separated clusters in the feature space.
- At-Risk and Lost customers showed overlap — indicating a spectrum of disengagement rather than a hard boundary.
- StandardScaler was applied before clustering to prevent Monetary (large values) from dominating Recency/Frequency.

Methodology

Step	Action	Detail
1	Data Collection	Online Retail II – 500,000+ transactions, 2009–2011
2	Data Cleaning	Removed nulls, returns (Qty < 0), zero-price records
3	RFM Calculation	Aggregated per CustomerID relative to snapshot date
4	Quintile Scoring	1–5 scores; Recency inverted (lower days = higher score)
5	Segmentation	Rule-based logic on R/F/M score combinations
6	K-Means Validation	Unsupervised ML (K=5) to validate rule-based clusters
7	Recommendations	Marketing actions mapped per segment

Conclusions & Strategic Recommendations

This analysis demonstrates how RFM segmentation can transform a flat customer database into a revenue-growth engine. The key strategic takeaways are:

- **Protect the top 25%.** Champions and Loyal Customers generate ~65% of revenue. Any churn in these groups has an outsized impact. Invest in loyalty programs, VIP perks, and early access offers.
- **Win back At-Risk customers fast.** These were once high-value buyers. A targeted win-back campaign with a personalized offer can recover 20–30% of this segment before they become fully lost.
- **Don't over-invest in Lost Customers.** One final reactivation email is worthwhile; beyond that, reallocate budget to nurturing New and Potential Loyalists.
- **Build a second-purchase funnel for New Customers.** The biggest retention drop-off happens between the 1st and 2nd purchase. A targeted welcome series can improve repeat-purchase rates by ~30%.
- **Monitor segment migration monthly.** RFM is most powerful when run regularly — tracking which customers move from Loyal to At-Risk allows proactive intervention before churn occurs.

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